QUESTION BANK (MCQ + Assertion–Reason)

SECTION A — Numbers, Encodings, Logic, Hardware

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1. Number Systems & Conversions (Binary, Octal, Decimal, Hex)

MCQs

1. **Which number system has base 16?**
   a) Binary
   b) Octal
   c) Decimal
   d) Hexadecimal
2. **Binary 10110_2 equals what in decimal?**
   a) 22
   b) 24
   c) 26
   d) 16
3. **The positional value of the rightmost digit is always:**
   a) $base^2$
   b) $base^0$
   c) $base^1$
   d) $base^3$
4. **Which method is used to convert decimal to binary?**
   a) Repeated subtraction
   b) Repeated division by 2
   c) Repeated division by 8
   d) Multiplication by 16

Assertion–Reason

5. **Assertion (A):** In positional number systems, each digit has a weight based on its position.
   Reason (R): The weight is given by base raised to a power.
   a) A and R true; R explains A
   b) A true, R true but no relation
   c) A true, R false
   d) A false, R true

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2. Binary Arithmetic, 1's & 2's Complement

MCQs

6. **1's complement of 101000_2 is:**
   a) 111000
   b) 010111
   c) 010111
   d) 010111_2
7. **2's complement is obtained by:**
   a) Flipping bits only
   b) Flipping bits + adding 1
   c) Adding 2
   d) Shifting left
8. **Which is true for 2's complement?**
   a) Two zeros exist
   b) Only one zero exists
   c) Cannot represent negatives
   d) Cannot do subtraction

Assertion–Reason

9. **Assertion (A):** 2's complement simplifies subtraction.
   Reason (R): It allows subtraction to be performed as addition.
   a) A and R true; R explains A
   b) A true; R true but unrelated
   c) A true; R false
   d) A false; R true

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3. Floating-Point, Mantissa–Exponent, Precision

MCQs

10. **Floating-point numbers store:**
    a) Sign only
    b) Mantissa + Exponent
    c) Exponent only
    d) Mantissa only
11. **Double precision uses how many bits?**
    a) 16
    b) 32
    c) 64
    d) 128
12. **Increasing mantissa size increases:**
    a) Range
    b) Precision
    c) Sign bit
    d) Overflow

4. Character Encodings — ASCII, Unicode, ISCII

MCQs

13. **ASCII supports how many characters?**
    a) 64
    b) 128
    c) 256
    d) 512
14. **Unicode is preferred because:**
    a) It is smaller
    b) It supports global languages
    c) It is faster
    d) It is only for English

Assertion–Reason

15. **Assertion (A):** ASCII cannot represent Bengali characters.
    Reason (R): ASCII uses only 7 bits.
    a) A and R true; R explains A
    b) A true; R true but unrelated
    c) A true; R false
    d) A false; R true

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5. Propositional Logic, WFF, Truth Tables

MCQs

16. **¬(P ∧ Q) is equivalent to:**
    a) ¬P ∧ ¬Q
    b) ¬P ∨ ¬Q
    c) P ∧ Q
    d) P ∨ Q
17. A formula true for all interpretations is:
    a) Satisfiable
    b) Unsatisfiable
    c) Valid
    d) Invalid

Assertion–Reason

18. **Assertion (A):** A WFF is a syntactically correct logical expression.
    Reason (R): WFFs must follow grammar rules.
    a) A and R true; R explains A
    b) A true, R true but unrelated
    c) A true, R false
    d) A false, R true

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6. Logic Gates & Adders

MCQs

19. **Universal gates are:**
    a) AND, OR
    b) NOT, XOR
    c) NAND, NOR
    d) XNOR, XOR
20. Which circuit adds two bits and produces sum + carry?
    a) Decoder
    b) Multiplexer
    c) Half Adder
    d) Full Adder

SECTION B — Java Programming Concepts

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7. Objects, Classes, JVM

MCQs

21. Objects are:
    a) Only data
    b) Only methods
    c) Data + methods
    d) Methods only
22. JVM is:
    a) Hardware CPU
    b) Software-based machine
    c) Operating system
    d) Database

Assertion–Reason

23. **Assertion (A):** Java programs are portable.
    Reason (R): JVM allows the same bytecode to run everywhere.

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8. Errors & Exceptions

MCQs

24. Which is a compile-time error?
    a) Divide by zero
    b) Missing semicolon
    c) NullPointerException
    d) ArrayIndexOutOfBounds
25.
26. Which keyword handles exceptions?
    a) try
    b) break
    c) new
    d) extends

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9. Types, Casting, Wrapper Classes

MCQs

26. Wrapper class of `int`:
    a) Integer
    b) Int
    c) Number
    d) Whole
27. Which is automatic type conversion?
    a) double → int
    b) int → double
    c) char → String
    d) long → byte

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10. Variables, Expressions, LHS–RHS

MCQs

28. In `x = x + 1`, the RHS `x` refers to:
    a) Memory address
    b) Value stored
    c) Garbage value
    d) Null
29. Which operator has highest precedence?
    a) +
    b) *
    c) =
    d) &&

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11. Statements, Scope, Loops

MCQs

30. A `break` statement exits from:
    a) Only switch
    b) Only loop
    c) Loop or switch
    d) Try block
31. A variable declared inside a block has:
    a) Global scope
    b) Method scope
    c) Block scope
    d) Class scope

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12. Methods & Constructors

MCQs

32. Constructors:
    a) Have no return type
    b) Return int
    c) Must be static
    d) Must be final
33. `this` refers to:
    a) Parent object
    b) Current object
    c) JVM
    d) Superclass

Assertion–Reason

34. **Assertion (A):** Constructors initialize objects.
    Reason (R): They run automatically when `new` is used.

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13. Arrays & Strings

MCQs

35. Array indexing in Java starts from:
    a) –1
    b) 0
    c) 1
    d) 2
36. The length of a string is obtained using:
    a) size()
    b) length
    c) length()
    d) count()

ANSWER KEY (1–2 Sentence Justifications)

1-d: Hex uses base 16.
2-a: $10110_2 = 22$.
3-b: Rightmost digit has weight base^0.
4-b: Decimal → Binary uses repeated division by 2.
5-a: Positional value = base^position.
6-b: Flip bits of 101000 → 010111.
7-b: 2's complement = 1's + 1.
8-b: 2's complement has only one zero.
9-a: Subtraction becomes addition using complements.
10-b: Floating point = mantissa + exponent.
11-c: Double precision uses 64 bits.
12-b: More mantissa bits increase precision.
13-b: ASCII uses 7 bits = 128 chars.
14-b: Unicode supports all world languages.
15-a: ASCII's 7-bit limit cannot encode Indian languages.
16-b: By De Morgan's law.
17-c: Valid = always true.
18-a: WFFs must follow syntax rules.
19-c: NAND & NOR alone can implement any circuit.
20-c: Half adder does sum + carry.
21-c: Objects store data + behaviour.
22-b: JVM is software.
23-a: Bytecode runs anywhere via JVM.
24-b: Missing semicolon is compile-time.
25-a: try handles exceptions.
26-a: Wrapper for int is Integer.
27-b: Widening conversion.
28-b: RHS refers to stored value.
29-b: * has higher precedence.
30-c: break exits loop or switch.
31-c: Inside a block → block scope.
32-a: Constructors have no return type.
33-b: this = current object.
34-a: They execute automatically during `new`.
35-b: Java arrays begin at index 0.
36-c: Strings use length().
